

The effect of inositol hexaphosphate on the expression of selected metalloproteinases and their tissue inhibitors in IL-1 β -stimulated colon cancer cells

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Introduction Matrix metalloproteinases (MMPs) have repeatedly been shown to play a very active role in extracellular matrix degradation associated with tumor invasion and metastasis. Tissue inhibitors of MMPs (TIMPs) are well-known for their ability to inhibit MMP activity thereby inhibiting malignant progression. Inositol hexaphosphate (IP6 phytic acid) has been recognized to have both preventive and therapeutic effects against various cancers including that of colon. In in vitro studies, IP6 has been demonstrated to inhibit cancer cell adhesion and migration. In the present study, the effect of IP6 on the expression of MMP and TIMP genes was evaluated in unstimulated and IL-1 β -stimulated colon cancer cell line Caco-2. **Materials and methods** Real-time QRT-PCR was used to validate the transcription level of selected MMP and TIMP genes in Caco-2 cells after treatment with 1 ng/ml of IL-1 β , 2.5 mM of IP6, and both for 6, 12, and 24 h. **Results** Stimulation of cells with IL-1 β only resulted in an overexpression of MMP and their TIMP mRNAs. A significant decrease in MMP-13, MMP-3, MMP-2, and TIMP-1 basal expression was achieved by IP6. IP6 was also an efficient downregulator of MMP-1, MMP-9, and TIMP-2 genes transcription stimulated by IL-1 β in 6 h lasting culture. After 12 h, IL-1 β -induced MMP-2 mRNA expression was significantly reduced by IP6. **Conclusion** Proinflammatory cytokine IL-1 β upregulates MMP and TIMP mRNAs expression in colon cancer epithelial cells Caco-2. IP6 (2.5 mM) influences constitutive expression of both MMP and TIMP genes and downregulates IL-1 β stimulated transcription of some of these genes. IP6 exerts its anti-metastatic activity through modulation of MMP and TIMP genes expression to prevent cancer cell migration and invasion.